Hierarchical Shell Manifold Theory: The Complete Mathematical Architecture: Grading, Reduction, Hierarchical Quotients, Relational Geometry, and Selection Boundaries

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October 2026
Mathematical structure monograph-style journal manuscript

Abstract

We present a consolidated mathematical architecture for Hierarchical Shell Manifold Theory (HSMT), separating its intrinsic operator structure, finite hierarchical models, continuum candidates, and physical-selection problem. The minimal datum is a separable Hilbert space with a strongly complete orthogonal grading and a densely defined reference operator. Under explicit domain and bounded-coupling hypotheses we establish self-adjoint stability, exact Schur–Feshbach reduction, memory dynamics, sector-leakage estimates, and coarse-graining of projection-valued measures. We formalize predictive quotients using strong lumpability of finite Markov generators, prove composition of exact quotients, and derive a finite-time error bound for approximate quotients. Bounded External Generative Complexity (BEGC) is recorded as a model-design hypothesis, not an established law. For relational geometry we construct contingency-table switch graphs, prove their energy positivity, analyze the exact three-state example, and show that the same relational Laplacian admits inequivalent norm-preserving dynamical completions. We distinguish graph, spectral, information, and operational dimensions; establish the affine dimension of an unconstrained transportation polytope; and prove a restricted positivity obstruction to Lorentzian signature under static Schur complementation. The reported autonomous hierarchy on the Johnson scheme, scalable symmetry-broken phases, higher-dimensional estimator coincidences, and elliptic continuum limits are included in an explicit technical-dependency register rather than presented as independently verified theorems without their underlying defining data. The result is a detailed, internally testable mathematical specification of HSMT and a clear boundary between available mathematics and an as-yet unselected physical theory.

Keywords: graded Hilbert spaces; Schur–Feshbach maps; lumpability; hierarchical quotient dynamics; BEGC; spectral graph geometry; transportation polytopes; physical selection.

1 Scope and claim standard

The distinction between mathematical possibility and physical identification is central to theories of emergence. A self-adjoint operator is not, by itself, an experimentally selected Hamiltonian; a graph metric is not automatically spatial distance; an index labeling direct-sum sectors is not necessarily an observed spatial dimension. HSMT is intended as a framework for keeping these identifications separate.

This paper has two purposes. First, it supplies a small, independently checkable mathematical foundation and a finite relational benchmark. Second, it records the status of more ambitious hierarchy-generation and geometric constructions without representing their unprovided proofs as established here. We distinguish four types of assertion throughout:

Established mathematical fact
A standard result cited from the literature or proved below.

Conditional construction
A result whose validity depends on a stated choice of graph, operator, dynamics, or other input.

Program claim
A result reported by the HSMT development program for which this article does not reproduce enough definitions and proof to certify it independently.

Open physical identification
A proposed connection to empirical physics for which no selection or measurement correspondence has yet been supplied.

This convention is not cosmetic: no claim of independent mathematical verification is made here for the program claims.

2 The graded operator framework

Definition 2.1 (Minimal graded core)**.** *A minimal graded core is a triple* $(\mathcal{H}, \{P_N\}_{N\in\mathcal{I}}, D)$*, where* $\mathcal{H}$ *is a separable complex Hilbert space,* $\mathcal{I}$ *is finite or countable,* P_N *are mutually orthogonal projections,*

$$P_N P_M = \delta_{NM} P_N, \qquad \sum_{N\in\mathcal{I}} P_N = \mathrm{I} \quad \textit{in the strong operator topology,} \tag{1}$$

and D *is a densely defined linear operator on* $\mathcal{H}$*. We write* $\mathcal{H}_N = P_N\mathcal{H}$*.*

The datum D is deliberately a *reference operator*, not necessarily the physical time-evolution generator. The definition does not assert that any particular N corresponds to a number of spatial dimensions or that a preferred physical grade exists. If block expressions $P_N D P_M$ are employed for unbounded D, their domains must be specified; they are not automatically operators defined on all of $\mathcal{H}_M$.

Proposition 2.2 (Bounded intergrade coupling)**.** *Suppose* D_0 *is self-adjoint on* $\mathrm{Dom}(D_0)$ *and* $T \in \mathcal{B}(\mathcal{H})$ *is self-adjoint. Then* $D_0 + T$ *is self-adjoint on* $\mathrm{Dom}(D_0)$*.*

Proof. A bounded T is D_0-bounded with relative bound zero. The conclusion is the bounded-perturbation case of the Kato–Rellich theorem [1, 2]. □

This theorem permits controlled intergrade hopping in a chosen model. Neither boundedness nor a nearest-neighbour hopping pattern follows from Eq. (1); both are independent assumptions.

2.1 Exact reduction and memory

Let $\mathcal{H} = \mathcal{H}_P \oplus \mathcal{H}_Q$, where P is an orthogonal projection and $Q = \mathrm{I} - P$. We consider a particularly transparent domain-controlled class:

$$H = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}, \qquad \mathrm{Dom}(H) = \mathrm{Dom}(A) \oplus \mathrm{Dom}(C), \tag{2}$$

with self-adjoint A, C and bounded $B : \mathcal{H}_Q \to \mathcal{H}_P$. These assumptions make H self-adjoint by Proposition 2.2.

Proposition 2.3 (Projected resolvent)**.** *For $z \in \rho(C)$, let*

$$S_P(z) = z - A - B(z - C)^{-1}B^*, \qquad \mathrm{Dom}(S_P(z)) = \mathrm{Dom}(A). \tag{3}$$

Then $z \in \rho(H)$ if and only if $S_P(z)$ has a bounded, everywhere-defined inverse from $\mathcal{H}_P$ to $\mathrm{Dom}(A)$ *(with its graph norm). In that case,*

$$P(z - H)^{-1}P\big|_{\mathcal{H}_P} = S_P(z)^{-1}. \tag{4}$$

Proof. For $(z - H)(u, v) = (f, g)$ the second block equation gives $v = (z - C)^{-1}(g + B^*u)$. Substitution in the first yields $S_P(z)u = f + B(z - C)^{-1}g$. This triangular block factorization has bounded off-diagonal factors and proves the stated inverse equivalence and identity. The result is a conventional Schur–Feshbach formula [3, 4, 5]. □

For an arbitrary $z \notin \mathbb{R}$, self-adjointness ensures $z \in \rho(C) \cap \rho(H)$, so Eq. (4) applies without additional spectral qualifications.

Writing $\psi(t) = (u(t), v(t))$ and taking $\mathrm{i}\partial_t\psi = H\psi$ gives

$$v(t) = e^{-\mathrm{i}Ct}v(0) - \mathrm{i}\int_0^t e^{-\mathrm{i}C(t-s)}B^*u(s)\,\mathrm{d}s, \tag{5}$$

when the evolution and initial data admit the displayed strong expressions (or in the usual mild sense). Therefore

$$\mathrm{i}\partial_t u(t) = Au(t) + Be^{-\mathrm{i}Ct}v(0) - \mathrm{i}\int_0^t Be^{-\mathrm{i}C(t-s)}B^*u(s)\,\mathrm{d}s. \tag{6}$$

The operator-valued memory factor is $Be^{-\mathrm{i}Ct}B^*$; the $-\mathrm{i}$ prefactor belongs to the integral in this convention. Equations Eqs. (4) and (6) are exact for an *already selected* projection P. They do not select which P is physically observable.

2.2 Sector persistence

For a bounded self-adjoint generator H and an orthogonal projection P, define the leakage operator $L_P = (\mathrm{I} - P)HP$ and $\varepsilon_P = \|L_P\|$. A reducing sector satisfies $L_P = 0$ (equivalently, $[H, P] = 0$ for self-adjoint bounded H). For $t \in \mathbb{R}$,

$$\|(\mathrm{I} - P)e^{-\mathrm{i}tH}P\| \le \min\{1, |t|\varepsilon_P\}. \tag{7}$$

Indeed, the derivative of $(\mathrm{I} - P)e^{-\mathrm{i}tH}P$ can be expressed through the Duhamel formula relative to the block-diagonal operator $PHP + (\mathrm{I} - P)H(\mathrm{I} - P)$, and integration gives the bound. For unbounded H such conclusions require further domain or bounded-commutator hypotheses.

Persistent sectors are thus meaningful after a generator is chosen, not an automatic consequence of grading.

3 Algebra of grade-resolved observables

3.1 Diagonal and off-diagonal decomposition

For bounded $X \in \mathcal{B}(\mathcal{H})$ define its diagonal grade restriction, whenever the strong sum is taken over a countable family,

$$\mathcal{E}(X) = \sum_{N \in \mathcal{I}} P_N X P_N. \tag{8}$$

The sum converges strongly because its partial sums act in mutually orthogonal ranges and $\|\mathcal{E}(X)\| \leq \|X\|$. It is a unital positive idempotent map and a bimodule map over the von Neumann algebra of grade-block-diagonal operators. No assumption identifies $\mathcal{E}$ with a physical decoherence channel.

Proposition 3.1 (Orthogonal block decomposition)**.** *For every $\psi \in \mathcal{H}$, $\sum_N \|P_N\psi\|^2 = \|\psi\|^2$. For bounded X, the expectation in a normalized vector ψ decomposes as*

$$\langle \psi, X\psi \rangle = \sum_N \langle P_N\psi, X P_N\psi \rangle + \sum_{N \neq M} \langle P_N\psi, X P_M\psi \rangle, \tag{9}$$

where the off-diagonal series is understood through the limit of finite grade cutoffs, not necessarily as an absolutely convergent double series.

Proof. The first identity follows from strong completeness of orthogonal projections. For $F \subset \mathcal{I}$ finite put $P_F = \sum_{N \in F} P_N$. Then $\langle P_F\psi, X P_F\psi \rangle \to \langle \psi, X\psi \rangle$ by norm convergence, and finite matrix multiplication yields the displayed expansion. □

3.2 Selected grades, reference operators, and generator freedom

A grade is a kinematic direct-sum label. A shell, in the broader HSMT vocabulary, is a regime of operational response that may extend across grades; its effective dimension is a separate scale-dependent diagnostic. A projection P_N may define a candidate sector, but grade selection requires an additional rule. The reference D defines spectral and coupling questions only after its operator domain and decomposition are fixed. A physical generator H need not equal D, nor need it preserve the grading. In particular, $D = D^*$ alone supplies neither a physical clock nor a privileged representation of measured observables.

3.3 Graded couplings and bounded hopping

Suppose $D_0 = \bigoplus_N D_N$ is self-adjoint on its direct-sum domain. Given a bounded symmetric block array T_{NM} defining a bounded self-adjoint T on the Hilbert direct sum, $D_0 + T$ is self-adjoint. A common concrete sufficient condition for a nearest-neighbour sequence indexed by $\mathbb{Z}$ is

$$(T\psi)_N = B_{N-1}\psi_{N-1} + B_N^*\psi_{N+1}, \qquad \sup_N \|B_N\| \leq b, \tag{10}$$

with orientations chosen consistently; then $\|T\| \leq 2b$. This realizes a banded coupling without deriving it from grading. A Gaussian profile $g_{NM} = \exp[-(N-M)^2/(2\sigma^2)]$ may instead be inserted as a modeling kernel; neither its width nor its normalization is selected by the minimal core.

4 Spectral inverses and response data

4.1 Finite-rank spectral response

If H is self-adjoint, $\phi \in \mathcal{H}$, and $z \notin \mathbb{R}$, the scalar diagonal resolvent

$$G_\phi(z) = \langle \phi, (z - H)^{-1}\phi \rangle = \int_{\mathbb{R}} \frac{1}{z - \lambda} \, \mathrm{d}\mu_\phi(\lambda) \tag{11}$$

is the Cauchy transform of the positive spectral measure $\mu_\phi(B) = \langle \phi, E_H(B)\phi \rangle$. This is a standard application of the spectral theorem, and defines a meaningful inverse problem only when a response function G_ϕ is supplied or observed.

Proposition 4.1 (Scalar self-energy reconstruction)**.** *Suppose a one-dimensional visible sector has bare level ω_0 and projected resolvent $G(z) = [z - \omega_0 - \Sigma(z)]^{-1}$ at a point where $G(z) \neq 0$. Then*

$$\Sigma(z) = z - \omega_0 - G(z)^{-1}. \tag{12}$$

For perturbed data $\widetilde{G} = G + \delta G$ with both factors nonzero,

$$\widetilde{\Sigma} - \Sigma = \frac{\delta G}{G(G + \delta G)}. \tag{13}$$

Proof. Both statements follow by exact algebraic inversion. The second identity shows explicitly that inversion becomes ill-conditioned near a zero of G. □

This inverse does not identify a unique microscopic bath: many realizations may have the same scalar spectral measure. Boundary values of Stieltjes transforms and proposed fractional response exponents require stated regularity, normalization, and limiting prescriptions. No particular spectral exponent is forced by the grade axioms.

4.2 Projected measurement and the POVM boundary

Given a normalized vector ψ and the grade projectors, the grade-outcome probabilities $p_N = \langle \psi, P_N \psi \rangle$ form a probability distribution. More general measurement statistics arise from a positive operator-valued measure (POVM) M through $\Pr(B \mid \rho) = \mathrm{Tr}(\rho M(B))$. A grade family is one possible projective measurement, not the unique canonical laboratory observable. Internal-fibre observables can be informationally complete inside a declared finite sector; this alone does not select one global packet, a canonical cross-grade localization observable, or an experimental preparation map.

5 Continuous parents and measured dimension

Proposition 5.1 (Coarse-graining a projection-valued measure)**.** *Let E be a projection-valued measure on a measurable space (X, Σ), and let $\{\Delta_N\}_{N \in \mathcal{I}}$ be a countable measurable partition of X up to an E-null set. Then $P_N = E(\Delta_N)$ satisfies Eq.* (1)*.*

Proof. Multiplicativity and countable additivity of the projection-valued measure give orthogonality and strong resolution of the identity. □

This shows that the graded core is compatible with a continuous parent. It does not show that a continuous parent is uniquely determined by a given grading, or that a particular partition is selected in nature.

A physical dimension estimator requires an operationally specified response $Q_N(\ell)$ and a scale range in which it can be measured. One possible definition is

$$d_N(\ell) = \frac{\mathrm{d}\log Q_N(\ell)}{\mathrm{d}\log \ell}, \qquad Q_N(\ell) > 0. \tag{14}$$

The dimension inferred depends on the choice of Q_N: spectral, volume-growth, information, and dynamical dimensions need not agree. In particular, $N = d_N$ is not a theorem. A monotonically relabeled continuous parameter also carries no preferred metric or distance scale until an independent ruler is introduced.

6 Generative hierarchy: definitions and verification boundary

An HSMT research direction introduces *Bounded External Generative Complexity* (BEGC) to constrain the complexity of responses among composite states. In the original research summary, the intended contrast was a shared bounded-complexity response rule versus an arbitrary table of pairwise relations. The source summary does not specify the encoding class, complexity measure, growth limit, or exact descent rules. Consequently, the verbal statement is insufficient to define a mathematical predicate for which a general theorem could be proved here.

A complete technical definition must at least identify: (i) a class $\mathscr{S}_n$ of exposed states; (ii) an admissible family of response maps R_n; (iii) a complexity functional $\mathsf{C}(R_n, \mathscr{S}_n)$; (iv) a precise uniform or asymptotic bound; and (v) the stability and predictive-congruence conditions that make a quotient an autonomous grade. Without these, the assertion that four microstates are necessary and sufficient for a minimal two-grade system cannot be independently evaluated.

6.1 Reported finite and scalable program results

The broader program records the following constructions; they are *not established in this paper*:

(i) A finite, symmetry-neutral construction based on $J(8,3)$ reportedly selects multiple simultaneous persistence-stable predictive quotients using a three-body rainbow statistic and a collision functional.

(ii) A scalable five-coordinate parent reportedly supports a symmetry-broken $3+2$ phase, approximate quotient closure, and finite representatives.

(iii) On associated fingerprint and contingency-switch structures, a path operator, response-locality criterion, several dimensional estimators, and an elliptic continuum limit have been reported.

To elevate these to theorems in a submission, a companion must state its configuration space and functional, prove the asserted minimizers and open phase region, quantify quotient defects, provide convergence hypotheses and limiting topology, and document the numerical or symbolic checks. Until such a companion is accessible, the claims are research targets/results as reported by the program, not independently reproduced contributions of this article.

7 Exact and approximate autonomous quotients

7.1 Finite state spaces and strong lumpability

Let Ω be a finite microstate set and let L be a continuous-time Markov generator in the row convention: $L_{xy} \geq 0$ for $x \neq y$, and $\sum_y L_{xy} = 0$. Fix a surjection $q : \Omega \to \bar{\Omega}$. Define the pullback $J : \mathbb{R}^{\bar{\Omega}} \to \mathbb{R}^{\Omega}$ by $(Jf)(x) = f(q(x))$. An autonomous quotient generator $\bar{L}$ satisfies

$$LJ = J\bar{L}. \tag{15}$$

This is the precise finite meaning of predictive closure for all block-constant observables.

Theorem 7.1 (Strong lumpability criterion)**.** *There exists a Markov generator $\bar{L}$ satisfying Eq. (15) if and only if, for every two microstates x, x' with $q(x) = q(x')$ and every macrostate b,*

$$\sum_{y:q(y)=b} L_{xy} = \sum_{y:q(y)=b} L_{x'y}. \tag{16}$$

When it exists, $\bar{L}$ is unique.

Proof. Apply LJ to each indicator $\mathbf{1}_{\{b\}}$ to obtain the necessity of the constant-block-sum property. Conversely define $\bar{L}_{ab}$ by the common value of the block sum for any x with $q(x) = a$. Nonnegative off-diagonal entries and zero row sums follow from the same properties of L. The indicator functions form a basis, so $LJ = J\bar{L}$. Surjectivity of q implies injectivity of J, hence uniqueness. □

Corollary 7.2 (Nested predictive autonomy)**.** *If $q_1 : \Omega \to \Omega_1$ and $q_2 : \Omega_1 \to \Omega_2$ are exact generator quotients with $LJ_1 = J_1L_1$ and $L_1J_2 = J_2L_2$, then $q_2 \circ q_1$ is an exact quotient with generator L_2.*

Proof. $L(J_1J_2) = J_1L_1J_2 = J_1J_2L_2$. □

7.2 Predictive congruence versus visual grouping

A partition is not automatically a predictive quotient. Strong lumpability requires the total rates toward every target block to agree among all states within a source block. A visually obvious grouping can fail Eq. (16); conversely a non-spatial partition can close exactly. A nested hierarchy therefore requires an explicit compatible family of partitions and an evolution law.

7.3 Approximate quotient defect

If the candidate macro-generator $\bar{L}$ has been chosen but exact closure fails, define the operator defect $R = LJ - J\bar{L}$. Equip both finite observable spaces with the supremum norm; for Markov semigroups $e^{tL}, e^{t\bar{L}}$, both are contractions.

Proposition 7.3 (Finite-time error bound)**.** *For $t \geq 0$,*

$$\|e^{tL}J - Je^{t\bar{L}}\|_{\infty\to\infty} \leq t\|R\|_{\infty\to\infty}. \tag{17}$$

Proof. The bounded-generator Duhamel formula gives $e^{tL}J - Je^{t\bar{L}} = \int_0^t e^{(t-s)L}Re^{s\bar{L}}\,\mathrm{d}s$. Contractivity bounds the integrand by $\|R\|$. □

The inequality controls a declared finite observation horizon. Uniform-in-time or thermodynamic-limit approximation requires additional mixing, spectral gap, or stability information; a small one-step defect alone is not sufficient.

7.4 Hierarchical phase criterion

A useful mathematical *definition*, not a derived law, is a family of models (Ω_k, L_k) with nested quotient maps $q_{k,j}$ and chosen reduced generators $\bar{L}_{k,j}$ whose defects $\epsilon_{k,j} = \|L_k J_{k,j} - J_{k,j}\bar{L}_{k,j}\|$ obey specified limits. Distinct grades should remain nontrivial and separated under a declared metric of quotient inequivalence; persistence and hereditary closure must be stated for the same mode of limit. These conditions separate a scalable autonomous phase from a finite isolated example.

8 Bounded external generative complexity as a conditional principle

The HSMT program uses *Bounded External Generative Complexity* (BEGC) to disallow an arbitrary independent parameter for each pairwise response. The following provides an explicit mathematical template, not a uniquely fixed formulation of the original BEGC program. Let Ω_k be finite, $n_k = |\Omega_k|$, and let F_k be a family of response kernels $R_k : \Omega_k \times \Omega_k \to \mathbb{R}$. Given a model complexity $c(R_k)$ (e.g., a bounded feature dimension plus circuit-description length), a BEGC-type restriction is

$$R_k(x, y) = \mathcal{R}_{\theta_k}(f_k(x), f_k(y)), \quad \dim f_k \le d_*, \quad c(\mathcal{R}_{\theta_k}) \le C_*, \tag{18}$$

where the constants do not grow with n_k. Additional regularity, encoding, and parameter-precision restrictions are essential: a single unbounded-precision real parameter could otherwise encode an arbitrary lookup table. The choice of description language and admissible preparation operations likewise matters. Thus Eq. (18) illustrates the finite-description intent without claiming that it defines a universal physical complexity invariant.

The original development reports a four-microstate minimum for two strict nontrivial nested grades, a six-state restricted operational witness, and an autonomous $J(8, 3)$ hierarchy. They are not inferred from Eq. (18). Their precise original complexity measure, quotient definitions, and certificates must be supplied in independent technical work before those results are usable as theorems of this manuscript. A generic $O(n^2)$ dense response table is disfavoured by fixed-description restrictions only after an encoding convention is fixed.

9 An exact finite relational benchmark

We now give a self-contained construction that does not rely on the preceding program claims.

Definition 9.1 (Contingency fibre and rectangle switch)**.** *For nonnegative integer margins* $r = (r_1, \dots, r_m)$ *and* $c = (c_1, \dots, c_n)$ *with* $\sum_i r_i = \sum_j c_j$*, define*

$$\mathcal{F}(r, c) = \left\{ X \in \mathbb{Z}_{\ge 0}^{m \times n} : \sum_j X_{ij} = r_i, \quad \sum_i X_{ij} = c_j \right\}. \tag{19}$$

A rectangle move changes four cells in distinct rows i, k *and columns* j, l *by* $\pm(E_{ij} + E_{kl} - E_{il} - E_{kj})$

whenever the resulting table remains nonnegative. The switch graph *has vertex set* $\mathcal{F}(r,c)$ *and an undirected edge for each admissible move.*

For two-way tables without structural zeros, elementary 2×2 moves connect each nonempty fixed-margin fibre; this is a standard Markov-basis fact, not a new HSMT theorem [6]. Constraints or quotient identifications can change the statement and require separate verification.

Give each undirected edge unit conductance and let A_G be the adjacency matrix and Δ_G the degree matrix. The combinatorial Laplacian

$$K = \Delta_G - A_G \tag{20}$$

is positive semidefinite because

$$\langle f, Kf\rangle = \sum_{\{x,y\}\in E(G)} |f(x)-f(y)|^2 \geq 0. \tag{21}$$

The construction is a relational operator on the state graph; by itself it is not an observed spatial Laplacian.

Example 9.2 (Three-state benchmark)**.** *For* 2×2 *margins* $r=c=(2,2)$*, every table is*

$$X_a = \begin{pmatrix} a & 2-a \\ 2-a & a \end{pmatrix}, \qquad a\in\{0,1,2\}. \tag{22}$$

One admissible switch increments or decrements a *by one. The switch graph is the path* P_3*, with Laplacian*

$$K = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \qquad \operatorname{spec}(K)=\{0,1,3\}. \tag{23}$$

The eigenvectors $(1,1,1)^T$*,* $(1,0,-1)^T$*, and* $(1,-2,1)^T$ *have the respective eigenvalues* $0,1,3$*.*

9.1 Different dynamics on the same relational graph

The static data K permit inequivalent dynamical completions. For $g>0$, one Schrödinger-type choice is

$$\mathrm{i}\dot{\psi} = gK\psi, \tag{24}$$

whose nonzero angular frequencies are g and $3g$. A second, reversible wave-type completion uses a pair of real three-component variables u, v:

$$\dot{u} = gK^{1/2}v, \qquad \dot{v} = -gK^{1/2}u, \tag{25}$$

where $K^{1/2}$ is the positive square root. Its nonzero frequencies are g and $\sqrt{3}g$. Equivalently, $\ddot{u} = -g^2Ku$. The positive norm $\|u\|^2+\|v\|^2$ is conserved. This wave realization is *not* graph-local in the same sense as incidence dynamics, because $K^{1/2}$ is generally dense.

A local incidence realization can instead be formulated on vertex and edge spaces. Choose an oriented incidence matrix $\partial:\mathbb{R}^{|E|}\to\mathbb{R}^{|V|}$ so that $K=\partial\partial^T$, and set

$$\mathbb{A} = \begin{pmatrix} 0 & g\partial \\ -g\partial^T & 0 \end{pmatrix}, \qquad \mathbb{A}^T=-\mathbb{A}, \qquad \mathbb{A}^2 = -g^2\begin{pmatrix} K & 0 \\ 0 & \partial^T\partial \end{pmatrix}. \tag{26}$$

Thus $(\dot{u},\dot{w}) = \mathbb{A}(u,w)$ conserves $\|u\|^2+\|w\|^2$ and reproduces the nonzero wave frequencies

$g, \sqrt{3}g$ on the vertex sector. Importantly, the identity $\mathbb{A}^2 = -g^2 K$ is a *vertex-block identity*, not an equality between matrices acting on different spaces.

Proposition 9.3 (Kinematic nonuniqueness)**.** *The positive graph Laplacian K of Example 9.2, even together with exact conservation of a positive quadratic norm under closed evolution, does not select a unique nonzero frequency ratio.*

Proof. Equation Eq. (24) is unitary and has ratio $3/1 = 3$. The skew-adjoint flow Eq. (26) conserves its positive Euclidean norm and has ratio $\sqrt{3}/1 = \sqrt{3}$. The two constructions use the same K and fulfill the stated conservation requirement, but have different spectra. Thus those premises alone cannot select between them. □

The ratios are invariant under one common rescaling of the time parameter within each model. Calling a ratio *clock-free* means only that an overall clock calibration cancels; any comparison to experimental frequencies still requires a physical preparation/measurement correspondence. The preceding is a mathematical discriminator, not evidence that either completion describes nature.

10 Relational geometry and candidate dimension diagnostics

10.1 Intrinsic combinatorial geometry

For a connected finite weighted graph $G = (V, E, w)$, define the path metric $d_G(x, y)$ as the minimum sum of positive edge lengths over paths from x to y. Define the weighted Laplacian

$$(K_w f)(x) = \sum_{y:\{x,y\}\in E} w_{xy}[f(x) - f(y)], \qquad w_{xy} = w_{yx} > 0. \tag{27}$$

The quadratic form $\frac{1}{2}\sum_{x,y} w_{xy}|f(x)-f(y)|^2$ is intrinsic to the weighted graph, but its interpretation as physical action or diffusion requires a generator choice. Graph automorphism invariance provides a useful consistency condition for any proposed kernel. It does not, without an added premise, select unique weights from connectivity alone.

10.2 Five inequivalent notions of dimension

For an operationally declared scale ℓ and a strictly positive response $Q(\ell)$, a local scaling index is

$$d_Q(\ell) = \frac{\mathrm{d}\log Q(\ell)}{\mathrm{d}\log \ell}, \tag{28}$$

when the derivative exists. One may define volume-growth dimension from ball counts, spectral dimension from return probabilities or heat traces, information dimension from coverings or entropies, a conductance dimension from resistance/conductivity scaling, and dynamical dimension from the dispersion of a selected generator. They need not coincide, and some are undefined or dominated by finite-size effects in small graphs. The label N in the graded Hilbert space is separate from every one of these estimators.

10.3 Transportation-polytope affine dimension

For fixed strictly positive margin vectors $r \in \mathbb{R}^m_{>0}$, $c \in \mathbb{R}^n_{>0}$ with the same total mass $T > 0$, define

$$\mathcal{T}(r, c) = \{X \in \mathbb{R}^{m\times n}_{\geq 0} : X\mathbf{1} = r,\ X^T\mathbf{1} = c\}. \tag{29}$$

Proposition 10.1 (Affine transportation rank)**.** *The affine hull of $\mathcal{T}(r, c)$ has dimension $(m-1)(n-1)$.*

Proof. The $m+n$ linear margin equations have one linear dependence, since both sets sum to the same total. They have rank $m+n-1$: one may solve uniquely for the last row and column in terms of the $(m-1)(n-1)$ leading entries. Thus the affine solution space has dimension $mn-(m+n-1) = (m-1)(n-1)$. The matrix $X_{ij} = r_i c_j/T$ is strictly positive, so the nonnegative inequalities do not reduce the dimension of the affine hull. □

This rank is a conventional affine dimension of the full positive-margin transportation polytope. It is *not* an emergent observed spatial dimension. Agreement of additional graph, spectral, information, and conductance estimators with this rank is a separate limit theorem requiring explicit edge weighting, geometry, scale regime, and proof. The reported C2 results are recorded as unverified-in-this-article dependencies.

10.4 Candidate continuum forms

A prospective convergence of rescaled discrete forms $\mathcal{E}_k(f)$ toward

$$\mathcal{E}_\infty(f) = \int_{\mathcal{T}} \nabla f(x)^T A(x) \nabla f(x)\, \mathrm{d}\mu(x), \qquad A(x) = A(x)^T \succeq 0, \tag{30}$$

would yield an elliptic diffusion only after specifying normalization, ambient measure, coefficient regularity, domains, boundary conditions, and a convergence notion such as Mosco convergence. A reflecting conormal boundary rule can arise naturally for appropriate excluded-crossing graph transitions but is not implied by the transportation constraints alone. Equation Eq. (30) states the target class, not a proved C2 limit.

11 A restricted signature obstruction

The relation between generated elliptic operators and possible Lorentzian descriptions should be stated with its assumptions visible.

Proposition 11.1 (Positive Schur complements preserve positivity)**.** *Let $M = \left(\begin{smallmatrix} A & B \\ B^T & C \end{smallmatrix}\right)$ be a real symmetric strictly positive-definite matrix, with C invertible. Then its Schur complement $A - BC^{-1}B^T$ is strictly positive-definite.*

Proof. For nonzero x, put $y = -C^{-1}B^T x$. Since $(x, y) \neq 0$,

$$x^T(A - BC^{-1}B^T)x = \begin{pmatrix} x \\ y \end{pmatrix}^T M \begin{pmatrix} x \\ y \end{pmatrix} > 0. \tag{31}$$

□

Consequently, a positive-definite *finite-dimensional principal-symbol quadratic form*, reduced through a positive block Schur complement satisfying these hypotheses, cannot acquire mixed Lorentzian signature by that reduction alone. This does *not* prove an impossibility for every projection, dynamical extension, frequency-dependent elimination, singular limit, or nonlocal effective theory. In particular, the incidence evolution in Eq. (26) adds a dynamical sector; it is not a static signature change of the sort covered by Proposition 11.1. Whether a particular continuum HSMT model supports a Lorentzian effective description remains an independent question.

12 What a physical-selection principle must supply

The core does not determine an operational theory. We make that limitation constructive by listing distinct missing maps. First, a *generator-selection map* must choose the physical law of evolution from the admissible operator families. Second, a *calibration map* must relate its parameter to physical clock readings. Third, a *sector/observable map* must specify what laboratory preparations, observables, and outcomes correspond to mathematical states and operators. Fourth, a *geometric correspondence* must identify which of the possible response quantities behaves as measured distance or dimension.

An additional conservation axiom, considered in the HSMT program, states that closed fundamental evolution preserves a canonical positive quadratic form. Within a declared linear finite-dimensional model this restricts a generator G through $G^T M + MG = 0$ for some selected $M > 0$; however there are generally many such G. Thus conservation alone is insufficient for unique selection, as Proposition 9.3 illustrates concretely.

The broader program has considered causal-order, operational-tomography, information-preservation, and laboratory-emulation premises. These investigations can be informative, but no claim that *all possible* external physical premises have been exhausted follows from a finite set of audits. It is appropriate to suspend one line of foundational search without treating that suspension as a no-go theorem about every future physical law.

13 Operational selection, conserved forms, and model discrimination

13.1 Positive-norm-preserving generators

Let $M = M^T > 0$ and let $\dot{x} = Gx$ on a finite real state space. Exact conservation of $E(x) = x^T Mx/2$ for every solution holds if and only if

$$G^T M + MG = 0. \tag{32}$$

Proposition 13.1 (Classification of linear positive-form conserving flows)**.** *The matrices satisfying Eq.* (32) *are exactly* $G = M^{-1/2} A M^{1/2}$ *with* $A^T = -A$*. In dimension at least two there are infinitely many such generators.*

Proof. Set $A = M^{1/2} G M^{-1/2}$. Then $A^T + A = M^{-1/2}(G^T M + MG)M^{-1/2}$. Thus conservation is equivalent to skew-symmetry. The space of real skew-symmetric matrices has dimension $d(d-1)/2$ in dimension d, providing infinitely many choices for $d \geq 2$. □

This result formalizes the scope and the insufficiency of the adopted HSMT reversibility premise DCP-0: conservation constrains a class of generators but does not single out the incidence realization, operational clock, or physical law.

13.2 Static tests versus dynamical tests

The finite graph P_3 fixes the unordered Laplacian eigenvalues $\{0, 1, 3\}$. It does not fix whether measured resonances scale with λ or $\sqrt{\lambda}$. The corresponding positive-frequency ratios 3 and $\sqrt{3}$ are calibration-invariant within their specified models. Laboratory measurements could distinguish two engineered realizations given common excitation and observation definitions.

Even a perfect measurement would not establish that the preferred realization is fundamental physics without a bridge from the emulator's measurable signals to a universal HSMT observable.

13.3 Physical correspondence package

A completed empirical theory would need at least a class of preparations $\mathcal{P}$, a probability assignment for laboratory outcomes $\Pr(o \mid p, t)$, a generator and its calibration to clock readings, and a correspondence mapping mathematical response scales to distances or dimensions measured experimentally. These are independent choices unless a theorem proves otherwise. The model may be mathematically self-consistent in their absence, but cannot make unique framework-level physical predictions.

14 Relationship to the broader HSMT program

The research record motivating this article includes proposed finite hierarchy phases, a scalable relational geometry construction, ring-exchange analogues, and a laboratory comparator. Their relationship to this framework is summarized in Table 1. In this article the switch graph and its spectra are completely explicit; the reported hierarchy phases and their continuum limits are not. A future technical companion should state all finite generators and exact energies, give reproducible parameter domains, identify finite-size and limit orders, and prove the asserted convergence results. The absence of such content from this article must not be conflated with a proof that the content cannot exist.

Table 1: Claim ledger for the revised framework article. "Proved here" includes standard statements reproved with explicit assumptions; it does not imply novelty.

Statement	Status in this article
Minimal grading and bounded perturbation	Definition and established theorem
Exact block resolvent and projected memory	Established fact, derived under stated domains
Continuous parent coarse-graining	Established fact, proved here
Finite 2×2 switch graph and P_3 spectrum	Explicit reproducible benchmark
Distinct reversible generator frequency ratios	Conditional models; nonuniqueness proved
Positive Schur complement signature obstruction	Proved under strict positivity assumptions
Four-state BEGC minimality, $J(8,3)$ hierarchy	Program claims; independent proof not included
Scalable phase and transportation-polytope continuum	Program claims; independent proof not included
Unique physical grade, time, measurement, dimension	Open physical identifications
P5 hardware measurement or physical validation	Not claimed; no bench results provided

15 Reproducibility and limitations

All numerical values presented as results *of this article* are finite exact quantities recoverable from Eq. (23). For example, the characteristic polynomial of K is $\lambda(\lambda-1)(\lambda-3)$; one may verify the results using symbolic arithmetic without external data. No new physical measurements, vendor SPICE simulations, or statistical datasets are claimed. A model-specific experiment comparing the two frequency ratios would test specified implementations, not independently establish HSMT as fundamental physics.

The principal limitation is intentional. This paper does not supply a novel physical law, a unique interpretation of the grade label, a derived $3+1$ spacetime, or a framework-level quantitative prediction. Nor does it provide complete proofs of the more ambitious hierarchy-generation and continuum claims summarized in Table 1. Those claims should not be submitted as self-standing theorems without accessible derivations. The mathematical contribution proved directly in this article is necessarily narrower than the entire program it describes. The expanded specification organizes the complete currently recorded architecture but does not silently promote unprovided program derivations to theorems.

16 Conclusion

HSMT can be formulated without conflating grading, spectral reduction, dimension measurement, and physical dynamics. Under explicit hypotheses its operator reductions are exact; a continuous parent can recover discrete grading; a finite contingency-switch system supplies a reproducible relational Laplacian; and two equally norm-preserving dynamic completions of the same graph yield distinct spectral ratios. A restricted Schur-complement argument explains why positive static reduction alone cannot provide an indefinite Lorentzian principal form, while leaving open other dynamical mechanisms.

The decisive unresolved issue is selection, not the mere availability of mathematical structures. Independent physical premises and operational correspondences would be needed to identify one dynamics and one experimentally meaningful interpretation. The reported BEGC and larger hierarchy phases are worthwhile targets for detailed companion publications, but their acceptance requires complete definitions and independently checkable proofs. This is a framework paper, not a claim that a physical theory has been uniquely derived.

A Reproducible finite calculations

For the graph P_3, write $K = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$. Direct expansion gives

$$\det(\lambda I - K) = \lambda(\lambda - 1)(\lambda - 3). \tag{33}$$

The orthogonal eigenvectors $v_0 = (1,1,1)$, $v_1 = (1,0,-1)$, and $v_3 = (1,-2,1)$ span $\mathbb{R}^3$. On the orthogonal complement of constants, Schrödinger evolution multiplies the v_1 and v_3 components by e^{-igt} and e^{-3igt}. Incidence evolution obeys $\ddot{u} = -g^2 K u$, with the same two modes oscillating at g and $\sqrt{3}g$. These identities completely reproduce the numeric spectral comparison reported in this article without code or fitted parameters.

B The claim-dependency and verification register

Subsystem	**Mathematical input required**	**Status and burden of proof**
Graded core	$\mathcal{H}$, projectors, domains, D	Defined here; operator consequences proved under assumptions
Exact projected dynamics	Bounded off-diagonal blocks, self-adjoint diagonals	Schur formula and memory equation derived here
Spectral inversion	A supplied scalar response G and bare level	Exact algebraic inversion, no microphysical uniqueness

Subsystem	Required input	Verification status
Continuous parent	PVM and specified measurable partition	Coarse-graining proved here; no natural partition selector
Predictive hierarchy	Finite Markov generator and state partitions	Exact and approximate quotient criteria proved here
BEGC finite classification	Fixed encoding and complexity notion; complete finite enumeration	Program-reported minimality; not independently certified here
C1 autonomous $J(8,3)$ phase	Explicit functional, collision term, parameter inequalities, minimizer proof	Program report; source certificates needed
C2-A scalable phase	Model sequence, Lyapunov function, mode of convergence	Program report; technical proof needed
C2-B intrinsic path operator	Fingerprint states, move costs, symmetry assumptions	Program report; independent graph derivation needed
C2-C response locality	Specified probes and Jensen–Shannon functional	Program report; precise inequalities needed
C2-D dimension estimators	Five definitions, scale regime, convergence proof	Program report; affine rank alone proved here
C2-E continuum diffusion	Rescaling, forms, measures, boundary conditions, convergence	Program report; not established by this article
Physical generator selection	Independent selection law and operational clock	Open
Physical spacetime / gauge / gravity	Measurement correspondences and dynamics	Open; not claimed
Experimental discriminator	Reproducible circuit models and measurements	Mathematical ratio proved; physical data not reported

C Minimal checklist for technical companion submissions

A mathematically complete HSMT hierarchy paper should state, for every claimed theorem: (i) the state set and topology, (ii) parameter space and uniformity assumptions, (iii) all operators, their domains and symmetries, (iv) the exact assertion with quantifiers, (v) proof or computational certificate and independent checks, (vi) order of finite-size and continuum limits, and (vii) the distinctions among existence, stability, selection, and experimental identification. The distinction between "one model exists" and "physics selects this model" must remain explicit throughout.

Statements and declarations

Funding. No funding was received for this work.
Competing interests. The author declares no competing interests.
Data and code availability. No new experimental datasets are reported. All finite benchmark matrices and exact spectra needed to reproduce the calculations established in this article are displayed explicitly. No claims of external reproduction of C1/C2 are made.
Generative AI assistance. OpenAI ChatGPT and xAI Grok were used as analytical and editorial assistants during drafting and review. Responsibility for verifying the mathematical and bibliographic content, approving the manuscript, and any eventual submission remains with the author.

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